

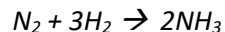
SEPTEMBER 17, 2019

At 25°C, an aqueous solution containing 35.0 wt% H_2SO_4 has a specific gravity of 1.2563. A quantity of the 35% solution is needed that contains 195.5 kg of H_2SO_4 .

- a) Calculate the required volume (L) of the solution using the given specific gravity.*
- b) Estimate the percentage error that would have resulted if pure-component specific gravities of H_2SO_4 (SG=1.8255) and water has been used for the calculation instead of the given specific gravity of the mixture.*

P2.43 Mixtures of hydrocarbon gases and air will burn only if the gas:air ratio is within the flammability limit. For example, propane gas-air will not ignite, even if exposed to flame, if the propane composition of the mixture is greater than 11.4 mol%. Calculate the pounds per hour of propane (C_3H_8) to mix with 100L/hr of air (at 1 atm pressure and 32°F) to just exceed the flammability limit and avoid ignition of the propane-air mixture. Air can be assumed to be 79 mol% N_2 and 21 mol% O_2 . The density of air or propane gas can be calculated from the ideal gas law.

(HW Review) P2.39 Nitrogen and hydrogen are fed at a 1:3 molar ratio to an ammonia synthesis reactor operating at 1440°R and 100 atm, where the following reaction takes place:



25% of the N_2 fed is converted to ammonia, and the reactor produces 1000 lbmol/h NH_3 . Calculate the volumetric gas feed rate to the reactor (ft^3/h), at the reactor temperature and pressure.